

## How the Proof Works

We start with a smooth, incompressible flow of finite energy and keep its viscosity fixed and positive. The problem is to prove that the flow remains smooth for all time. Its velocity  $u$  and pressure  $p$  satisfy the three-dimensional Navier–Stokes equation:

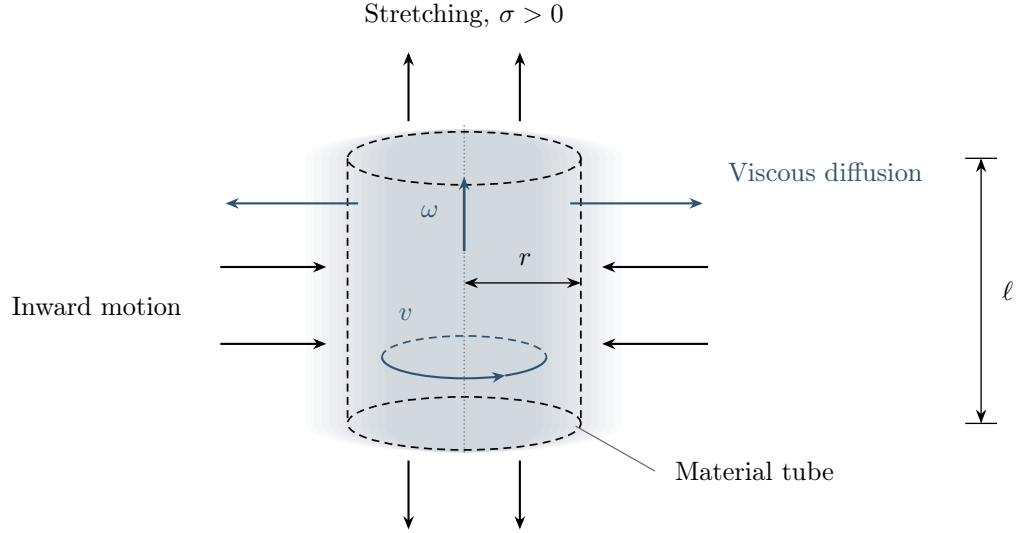
$$\underbrace{\partial_t u}_{\text{local acceleration}} + \underbrace{(u \cdot \nabla)u}_{\text{nonlinear transport}} = \underbrace{-\nabla p}_{\text{pressure}} + \underbrace{\nu \Delta u}_{\text{viscous diffusion}}, \quad \nabla \cdot u = 0.$$

The hope is simple: viscosity smooths the fluid faster than nonlinear transport can sharpen it.

Consider the case of a vortex within the fluid. Its vorticity  $\omega = \nabla \times u$  satisfies

$$\partial_t \omega + (u \cdot \nabla) \omega = (\omega \cdot \nabla) u + \nu \Delta \omega.$$

Strain within the fluid's flow can naturally stretch a vortex. Incompressibility forces a tube of fluid particles along its core to narrow as it lengthens.



For a thin, locally circular tube with length  $\ell$ , radius  $r$ , and positive axial strain  $\sigma$ ,

$$\dot{\ell} = \sigma \ell > 0, \quad r \propto \ell^{-1/2}.$$

If circulation  $\Gamma$  is retained through that narrowing, the mean swirl speed  $v$  increases:

$$v = \frac{|\Gamma|}{2\pi r}.$$

Viscosity diffuses momentum between neighbouring regions of fluid, reducing velocity differences and spreading rotation through the surrounding fluid. Over an elapsed time  $\tau$ , diffusion reaches a distance of order

$$r_{\text{diff}} \sim \sqrt{\nu \tau}.$$

The vortex core can spread beyond the narrowing material tube, weakening the circulation enclosed by it. Faster swirl depends on that circulation remaining strong enough as the radius shrinks.

A narrower core gives viscosity less distance to act across, but the growing velocity can also sharpen the flow more quickly. To compare these effects at progressively smaller scales, we rescale the flow by  $\lambda > 1$ :

$$\begin{aligned} u_\lambda(x, t) &= \lambda u(\lambda x, \lambda^2 t), \\ p_\lambda(x, t) &= \lambda^2 p(\lambda x, \lambda^2 t). \end{aligned}$$

Lengths shrink by  $\lambda$ , velocities grow by  $\lambda$ , and times shrink by  $\lambda^2$ . Every term in the momentum equation gains a factor  $\lambda^3$ , leaving the rescaled flow governed by the original equation at viscosity  $\nu$ . Smaller scales alone give viscosity no advantage over transport.

To measure the concentration, we need to count spatial variation as well as the size of the velocity. The Sobolev index  $s$  sets how strongly we count that variation: integer values count spatial derivatives in  $L^2$ , while fractional values weight frequency  $\xi$  by  $|\xi|^s$ . Velocity contributes  $\lambda$ , differentiation  $\lambda^s$ , and the square root of the volume  $\lambda^{-3/2}$ , giving

$$\|u_\lambda(\cdot, t)\|_{\dot{H}^s} = \lambda^{s-1/2} \|u(\cdot, \lambda^2 t)\|_{\dot{H}^s}.$$

Below  $s = \frac{1}{2}$ , concentration makes this norm smaller; above  $s = \frac{1}{2}$ , it makes it larger. At exactly  $s = \frac{1}{2}$ , the increased variation offsets the shrinking volume. This critical height lets us compare concentration without the measurement changing simply because we changed scale.

Energy sits below that height. For  $E = \frac{1}{2} \|u\|_2^2$ ,

$$E[u_\lambda](t) = \lambda^{-1} E[u](\lambda^2 t).$$

The velocity grows while total energy falls: the faster motion occupies less volume. An energy bound alone cannot exclude this concentration.

Nor does concentration necessarily run out of time. At width  $r$  and velocity scale  $U$ , transport acts over a time of order  $r/U$ , and viscosity over  $r^2/\nu$ . Both shorten by  $\lambda^2$  under the rescaling. If a flow could repeat these narrowing stages, their durations would fit into a finite total:

$$\tau_j = \lambda^{-2j} \tau_0, \quad \sum_{j=0}^{\infty} \tau_j = \frac{\tau_0}{1 - \lambda^{-2}} < \infty.$$

In this hypothetical Zeno cascade, infinitely many stages accumulate at a finite time  $T_*$ . There is no smallest width or final stage of acceleration. The scaling allows this picture; the regularity problem is whether the fluid can actually carry it out.

For a finite-time singularity, velocity would have to blow up. That requires acceleration to blow up, then jerk, and so on, and so on. These changes in velocity are governed by the Navier–Stokes equation. Differentiating it shows what each further temporal order involves:

$$\begin{aligned} \partial_t u &= \nu \Delta u - (u \cdot \nabla) u - \nabla p, \\ \partial_t^2 u &= \nu \Delta \partial_t u - (\partial_t u \cdot \nabla) u \\ &\quad - (u \cdot \nabla) \partial_t u - \nabla \partial_t p, \\ \partial_t^3 u &= \nu \Delta \partial_t^2 u - (\partial_t^2 u \cdot \nabla) u \\ &\quad - 2(\partial_t u \cdot \nabla) \partial_t u - (u \cdot \nabla) \partial_t^2 u - \nabla \partial_t^2 p. \end{aligned}$$

Each time derivative splits the transport product and applies a Laplacian to the preceding response. Substituting the first equation into the second exposes  $\Delta^2 u$ ; the next substitution exposes  $\Delta^3 u$ . Higher temporal orders reach further into the spatial derivatives, with the nonlinear terms carried along:

$$\partial_t^{n+1} u = \nu \Delta \partial_t^n u - \sum_{a=0}^n \binom{n}{a} (\partial_t^a u \cdot \nabla) \partial_t^{n-a} u - \nabla \partial_t^n p.$$

At the critical height, this relation exposes the spatial regularity behind those temporal demands. Write  $E_s = \frac{1}{2} \|\Lambda^s u\|_2^2$ , where  $\Lambda = (-\Delta)^{1/2}$ , and let  $H = E_{1/2}$ . Differentiating  $E_s$  and integrating the viscous term by parts moves one spatial derivative onto each factor:

$$E'_s = -2\nu E_{s+1} + \mathcal{P}_s,$$

with the transport and pressure contribution

$$\mathcal{P}_s = -\operatorname{Re} \langle \Lambda^s u, \Lambda^s ((u \cdot \nabla) u + \nabla p) \rangle_{L^2}.$$

Starting at  $s = \frac{1}{2}$ , each further time derivative exposes the next spatial energy:

$$\begin{aligned} H' &= -2\nu E_{3/2} + \mathcal{P}_{1/2}, \\ H'' &= 4\nu^2 E_{5/2} - 2\nu \mathcal{P}_{3/2} + \partial_t \mathcal{P}_{1/2}. \end{aligned}$$

Continuing through  $H'''$ ,  $H^{(4)}$ ,  $\dots$  reaches  $E_{7/2}, E_{9/2}, \dots$ , together with the differentiated nonlinear contributions.

Those spatial orders are enough for smoothness. Any chosen number  $k$  of continuous spatial derivatives needs only a finite Sobolev height:

$$m > k + \frac{3}{2} \implies H^m(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3).$$

Every choice of  $k$  is covered by a sufficiently high half-integer order. With finite energy, their intersection is

$$L^2 \cap \bigcap_{n \geq 0} \dot{H}^{n+1/2} = \bigcap_{m \geq 0} H^m = H^\infty \hookrightarrow C^\infty.$$

The temporal requirements of a singularity have led us into the spatial requirements of smoothness. If these orders remain controlled together through  $T_*$ , their limiting velocity is smooth enough to start the flow again.

If a singularity were to form, we would expect the opposite: smoothness at every earlier instant, but an unbounded fixed Sobolev norm as  $t$  approaches  $T_*$ . The temporal responses would become unbounded too, even while total energy remained finite. Having all finite derivatives before  $T_*$  would not give us the control needed to carry them through it.

That control must come from the initial velocity  $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$  under the equation. Incompressibility fixes pressure through  $-\Delta p = \partial_i \partial_j (u_i u_j)$ , with its decay normalization. Writing  $V_n = \partial_t^n u$  and  $Q_n = \partial_t^n p$ , the velocity and pressure responses obey

$$\begin{aligned} V_{n+1} &= \nu \Delta V_n - \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n, \\ Q_n &= R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j}. \end{aligned}$$

Here  $R_i$  are the Riesz transforms and repeated spatial indices are summed. Starting with  $V_0 = u_0$  at  $t = 0$ , these relations determine each subsequent response. Spatial differentiation fills out each temporal row. The resulting collection of velocity and pressure derivatives is the mixed jet generated by the datum.

We can retain any finite number of temporal rows and spatial orders. Choosing a temporal depth  $N$  and spatial height  $M$  gives a rectangle  $\mathcal{R}_{N,M}[u_0; T_*]$ . The datum-generated rectangle theorem, Theorem 4.6, controls its adjacent temporal rows throughout the approach to  $T_*$ :

$$\sup_{0 < T < T_*} \sum_{n=0}^N \left( \|V_n\|_{L^2(0,T;H^{M+1})}^2 + \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2 \right) < \infty.$$

The bound can depend on the orders  $N, M$ , but remains finite as  $T$  approaches  $T_*$ . Since  $V_{n+1} = \partial_t V_n$ , the adjacent rows control both a response and its change in time. The spaces  $H^{M+1}$  and  $H^{M-1}$  pair across  $H^M$ ; their time-integrable product gives a continuous endpoint trace by the Hilbert-triple theorem:

$$V_n \in C([0, T_*]; H^M).$$

A larger rectangle retains the derivatives already present. On each overlap, both rectangles contain derivatives of the original velocity and pressure, so their entries and endpoint traces agree. We can increase temporal depth and spatial height independently without losing that agreement. Their compatible intersection gives

$$\begin{aligned} \mathcal{I}[u_0] &:= (\mathcal{R}_{N,M}[u_0; T_*])_{N,M \geq 0}, \\ u &\in \bigcap_{r,m \geq 0} H^r(0, T_*; H^m(\mathbb{R}^3)). \end{aligned}$$

Every finite request for derivatives fits inside one of these rectangles. In particular, the velocity reaches  $T_*$  in every spatial Sobolev space, and its traces agree as one limiting velocity:

$$\begin{aligned} u_* &= \lim_{t \uparrow T_*} u(t) \quad \text{in every } H^m, \\ u_* &\in H_\sigma^\infty \hookrightarrow C^\infty. \end{aligned}$$

Smooth local existence starts the flow again from  $u_*$ , and uniqueness joins it to the original solution. It continues beyond its supposed maximal time. A finite  $T_*$  is impossible, so  $T_* = \infty$ .

The constructed intersection now supplies finite estimates at any chosen orders. For any finite time  $T$ , temporal order  $r$ , spatial order  $k$ , and  $2 \leq p \leq \infty$ , choose  $m > k + \frac{3}{2} - \frac{3}{p}$ . Sobolev embedding gives

$$\partial_t^r u \in C([0, T]; H^m) \hookrightarrow L^\infty(0, T; W^{k,p}).$$

In particular, taking  $r = 0$ ,  $k = 1$ ,  $p = \infty$ , and  $m = 3$  controls the total strain over any finite time:

$$\int_0^T \|\nabla u(t)\|_\infty dt \leq T \sup_{0 \leq t \leq T} \|\nabla u(t)\|_\infty < \infty.$$

For the locally circular material tube at the start,  $\dot{r}/r = -\sigma/2$  and  $\sigma \leq \|\nabla u\|_\infty$ . Hence

$$r(t) \geq r(0) \exp\left(-\frac{1}{2} \int_0^t \|\nabla u(s)\|_\infty ds\right) > 0.$$

Only a finite amount of stretching can accumulate in a finite time. The tube cannot be squeezed to zero radius there.